NU MOA
School of Information Technology
BS Information Technology

PRE-PROPOSAL CAPSTONE TOPIC FORM

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TITLE OF PROPOSED CAPSTONE PROJECT *The title should briefly and simply reflect the main topic of the project.*	
1. **Nexora: A Learning Management with Learning Experience Platform Features for Targeted Student Intervention for Gat Andres Bonifacio High School**	
SUSTAINABLE DEVELOPMENT GOAL (SDG)	SDG 4 – Quality Education SDG 9 – Industry, Innovation, and Infrastructure
CHOOSE SPECIALIZATION:	☒Mobile and Web Applications (MWA)
	☐ Multimedia Arts and Animation (MAA)
	☐ General Course
CAPSTONE PROJECT DISCIPLINE:	Mobile and Web Software Development

1. PROJECT CONTEXT

Please state the rationale of your proposed Capstone Project

Learning Management Systems (LMS) are widely used in high schools to support lesson delivery, assessment, and class management. However, most existing LMS platforms primarily focus on content distribution and administrative functions, offering limited support for targeted academic intervention for students who demonstrate learning difficulties.

At Gat Andres Bonifacio High School, students across all high school grade levels exhibit varying levels of mastery in different subjects. While teachers can identify struggling students through assessments, providing structured and timely intervention remains challenging due to limited instructional time, large class sizes, and the absence of an integrated intervention-focused system.

This project proposes the development of Nexora, a Learning Management System with Learning Experience Platform (LXP) features designed to support targeted student intervention for the entire high school population of Gat Andres Bonifacio High School. The LMS serves as the primary platform for managing classes, lessons, and assessments for all students, while the LXP functions as a performance-based intervention component.

Access to the LXP is restricted to selected students who demonstrate learning gaps, such as those obtaining scores below the defined performance threshold (e.g., below 60%). The LXP is directly connected to the LMS to ensure continuity of instruction, allowing students to revisit previous lessons and assessments as part of remediation. Automated instructional formulas generated from uploaded modules guide learning paths, while teachers retain full control over instruction and intervention through the LMS.

2. OBJECTIVES

State the general objective and at least 5 specific objectives of the study

General Objective:
To design and develop a web and mobile-based Learning Management System (LMS) integrated with a Learning Experience Platform (LXP) that supports all subjects across all grade levels, provides targeted learning intervention for selected students who demonstrate learning gaps, and assists teachers through automated instructional support, performance-based intervention, and system evaluation.

Specific Objectives:

1. To develop an LMS that allows teachers and administrators to manage classes, students, lessons, assessments, and learning progress effectively, including support for automatic formatting and extraction of uploaded learning modules.

2. To integrate an LXP within the LMS that is accessible only to selected students who demonstrate learning gaps (e.g., scores below 60%), ensuring targeted and effective intervention.

3. To design and integrate an AI NPC mentor that assists students by explaining mistakes, providing hints, and recommending remedial activities.

4. To conduct a system evaluation to assess the usability, functionality, performance, and user satisfaction of the LMS–LXP integration from the perspectives of teachers and selected students.

5. To provide teacher support features that assist educators in identifying learning gaps, managing intervention activities through the LMS-linked LXP, and reducing instructional workload through automated content processing.

3. SIGNIFICANCE OF THE STUDY
Discuss the importance of the project once developed

- For Students, the system provides structured academic support through targeted intervention. Selected students are given access to the LXP, allowing them to review previous lessons, revisit assessments, and follow guided remediation paths based on their performance. This approach supports mastery of learning competencies and reduces learning gaps across subjects.

- For Teachers, Nexora offers clear instructional benefits. Teachers can easily upload learning modules through automatic content extraction, identify students who require intervention, and manage LXP activities directly through the LMS. Automated instructional formulas assist in recommending remediation without replacing teacher judgment, helping reduce workload while improving intervention effectiveness.

- For the School, the system promotes data-informed academic intervention while maintaining a unified learning environment. The integration of LXP within the LMS allows structured remediation without disrupting regular instruction. Optional integration of intervention results into the LMS provides reference data for academic support, without functioning as official class records.

- For Future Researchers and Developers, the project serves as a reference for designing LMS–LXP systems that support performance-based intervention, controlled access, and system evaluation within a high school setting.

4. SCOPE AND DELIMITATION

Refers to the scope and inclusive frame/s of references as well as delimitation/s of the study

Scope

• The system supports all high school grade levels and subjects at Gat Andres Bonifacio High School through an LMS.
• The Learning Experience Platform (LXP) is designed for targeted intervention and is accessible only to selected students based on assessment performance (e.g., scores below 60%).
• Students are allowed to revisit previous lessons and assessments as part of the intervention process.
• Teachers access, manage, and monitor LXP activities exclusively through the LMS.
• The system includes automated instructional formulas generated from uploaded learning modules to support intervention.
• Teachers can upload modules with ease through automatic content extraction and formatting features.
• The platform is accessible via web and mobile devices.
• System evaluation is included to assess usability, functionality, performance, and user satisfaction.

Delimitation

• The system does not replace classroom instruction or teacher decision-making.
• Full automation of grading and official class record management is excluded.
• The LXP functions solely as an intervention component and is not intended for general student use.
• Integration of intervention results into the LMS is optional and for reference purposes only.
• Advanced predictive analytics and large-scale AI model training are not included.
• Deployment beyond Gat Andres Bonifacio High School is outside the scope of the study.
• Offline functionality and third-party content integrations are excluded.

5. PROJECT DESIGN / DEVELOPMENT PLAN

- Program Specification
- • System Architecture: Web and Mobile-Based Client–Server Model
- • Core Modules: User Management, LMS Core, LXP Engine, AI Mentor Module, Analytics Dashboard
- • Data Design: Relational Database (Student Records, Progress Logs, Assessments, AI Feedback History)
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- Software Specification
- • Frontend: Web and Mobile Frameworks
- • Backend: Node.js / Express.js (subject to change)
- • Database: PostgreSQL
- • AI Logic: Rule-based + Lightweight AI Integration, Mistral 7B for the LLM, but can be changed during recommendation

- Hardware Specification
- • Standard desktop or laptop computers for development
- • Mobile devices for testing
- • School-compatible server or cloud deployment environment

6. REFERENCES
Use IEEE citation and referencing style

AI Mentoring Benefits

AI-assisted mentors in LMS provide instant hints, adaptive difficulty, and feedback without replacing teachers, aligning with DepEd curricula for Grade 9 math. Studies show AI NPCs enhance comprehension by guiding problem-solving and addressing gaps dynamically. Combined with LMS, these features yield better outcomes in resource-limited environments.

M. Fitrah *et al.*, “Impact of learning management systems and digital skills on TPACK development among pre-service mathematics Teachers,” *Qubahan Academic Journal*, vol. 5, no. 1, pp. 504–518, Mar. 2025, doi: 10.48161/qaj.v5n1a1392.

The Springer article "Learning Experience Platforms (LXPs): A Primer on a Rapidly Evolving Field" examines how LXPs advance beyond traditional LMS by integrating AI-driven personalization, content curation, and real-time analytics to enhance learner engagement and outcomes. This directly aligns with Nexora, your LMS augmented with LXP features for targeted interventions at Gat Andres Bonifacio High School, as both emphasize data-informed strategies like progress tracking and early identification of at-risk students to deliver adaptive support in resource-constrained educational settings. By blending LMS structure with LXP's dynamic, learner-centric tools, Nexora embodies the article's vision for hybrid platforms that bridge learning gaps through intuitive, school-specific interventions.

G. Albano, F. Arzarello, and U. D. Iacono, “Digital inquiry through games,” *Technology Knowledge and Learning*, vol. 26, no. 3, pp. 577–595, Aug. 2020, doi: 10.1007/s10758-020-09459-1.

Problem-based learning approaches have been shown to enhance students’ mathematical connection ability and overall performance in mathematics. A study presented in the *Journal of Physics:*

Conference Series reported that students taught using a problem-based learning method with structured review activities achieved higher post-test scores and greater gains in mathematical connection ability than those receiving conventional instruction, indicating improved conceptual links and problem-solving skills. This supports the notion that interactive, structured learning strategies can effectively improve mathematics outcomes when embedded within digital or blended environments such as a web- and mobile-based LMS.

P. N. Malasari, H. Nindiasari, and Jaenudin, “A Development of Mathematical Connecting Ability of Students in Junior High School through a Problem-Based Learning with Course Review Horay Method,” *Journal of Physics Conference Series*, vol. 812, p. 012025, Feb. 2017, doi: 10.1088/1742-6596/812/1/012025.

Implementing technology in mathematics instruction has been found to support student engagement, reduce anxiety, and deepen conceptual understanding. A literature review on the effects of technology in high school mathematics classrooms showed that technology tools can increase student engagement, improve accuracy in computation tasks, create a less-anxious learning environment, and motivate learners toward deeper understanding of mathematics content.

D. Murphy, “A literature review: The Effect of Implementing Technology in a High School Mathematics Classroom,” *International Journal of Research in Education and Science*, vol. 2–2, pp. 295–299, season-02 2016.

A study on the use of digital escape game intervention examined its impact on secondary school students’ mathematics performance and affective constructs. The research with 32 students found that participation in the digital escape game significantly improved mathematics performance across diverse ability levels, reinforced interest and enjoyment in mathematics, and encouraged collaborative learning and communication while solving mathematical tasks.

D. N. N. P. A. Bakar, M. Shahrill, and Y. F. Zakariya, “Digital Escape game and students’ learning outcomes in Mathematics: experience from Brunei,” *SAGE Open*, vol. 13, no. 4, Oct. 2023, doi: 10.1177/21582440231216838.

ALL MODULES

1. User Management Module

- User Account Creation (Student, Teacher, Admin)

• Role-Based Access Control (RBAC)

• User Authentication (Login / Logout)

• Account Status Management (Active, Inactive)

• User Profile Management

• Password Encryption and Security

• User Data Validation

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2. Role & Access Control Module

- Role Definition (Student, Teacher, Administrator)

• Role Assignment to Users

• Permission Enforcement per Role

• Restricted Access to LXP (Performance-Based)

• Admin-Controlled Role Management

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3. Student Profile Module

- Student Personal Information

• Grade Level Assignment

• Subject Enrollment Records

• Learning Progress Tracking

• Assessment History

• Intervention Eligibility Status

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4. Teacher Profile Module

- Teacher Personal Information
- Class record and monitoring module

• Subject Handling Assignment

• Class Management Access

• Uploaded Learning Module Ownership

• Intervention Monitoring Access

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5. Class & Subject Management Module (LMS Core)

- Class Creation and Management

• Subject Assignment per Class

• Student Enrollment per Class

• Teacher–Class Mapping

• Academic Structure Management (Grade Level, Sections)

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6. Learning Content Management Module (LMS)

- Learning Module Upload (Documents / Files)

• Automatic Content Extraction

• Automatic Formatting of Uploaded Modules

• Lesson Organization per Subject

• Lesson Version Management

• Content Reuse for Intervention

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7. Assessment Management Module

- Assessment Creation (Quizzes, Tests)

• Assessment Scheduling

• Score Recording

• Performance Threshold Definition (e.g., below 60%)

• Assessment Review Access

• Assessment History Tracking

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8. Performance Tracking & Evaluation Module

- Student Performance Monitoring

• Identification of Learning Gaps

• Score-Based Performance Analysis

• Progress Comparison (Before & After Intervention)

• Performance Logs Storage

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9. Learning Experience Platform (LXP) Module

- Controlled Access for Selected Students Only

• Performance-Based Student Eligibility

• Remedial Learning Path Access

• Review of Previous Lessons

• Review of Previous Assessments

• Guided Remediation Activities

• LMS–LXP Content Continuity

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10. Intervention Management Module

- Automatic Intervention Trigger (Below Threshold)

• Teacher-Managed Intervention Activities

• Assignment of Remedial Tasks

• Monitoring of Intervention Progress

• Optional Integration of Intervention Results into LMS

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11. AI Mentor (AI NPC) Module

- AI-Assisted Explanation of Mistakes

• Step-by-Step Hints and Guidance

• Recommendation of Remedial Activities

• Context-Based Feedback from Assessments

• AI Feedback History Logging

• Teacher-Controlled AI Assistance Scope

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12. Instructional Support Module (Teacher Support)

- Automated Instructional Formula Generation

• Learning Gap Detection Assistance

• Suggested Remediation Paths

• Reduced Manual Workload for Teachers

• Teacher Override and Decision Control

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13. Analytics & Dashboard Module

- Student Progress Visualization

• Intervention Effectiveness Overview

• Class Performance Summary

• Teacher Dashboard View

• Admin System Overview Dashboard

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14. Reporting Module

- Student Master List Report

• Class Enrollment Report

• Student Performance Report

• Intervention Participation Report

• Assessment Summary Report

• System Usage Report (Teachers & Students)

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15. System Evaluation Module

- Usability Evaluation Tools

• Functionality Assessment

• Performance Evaluation

• User Satisfaction Feedback (Teachers & Students)

• Evaluation Data Storage

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16. Security & Data Management Module

- Secure Data Storage (PostgreSQL)

• Encrypted Credentials

• Access Control Enforcement

• Data Integrity Validation

• Activity Logging

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17. Web & Mobile Access Module

- Web-Based System Access

• Mobile-Friendly Interface

• Responsive Design

• Cross-Platform Compatibility